

VPN Protocols

FIG_000 · guide v1.0 · 2026 · open reference · 9 sections

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How two networks build an encrypted tunnel across the untrusted internet – IPsec's two-phase handshake, WireGuard's modern minimalism, and the practical trade-offs between site-to-site and remote-access designs.

```
$ read section_01 --start
```

WHAT'S INSIDE

SECTION	TOPIC
1 – 3	Why VPNs exist, and IPsec's two-phase handshake in depth
4 – 5	WireGuard's modern design, and a worked handshake walkthrough
6 – 7	Split vs full tunneling, and site-to-site vs remote-access designs
8 – 9	A worked site-to-site configuration example, and reference

How to use this guide: read start to finish for the full picture, or jump to section 4 if you just want WireGuard specifically. Every diagram is numbered and referenced directly in the surrounding text.

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This guide is designed to be read section by section, with diagrams positioned next to the concepts they illustrate.

What Is a VPN & Why It Exists

1.1 AN ENCRYPTED TUNNEL ACROSS AN UNTRUSTED NETWORK

A **VPN (Virtual Private Network)** creates an encrypted tunnel across an untrusted network – almost always the public internet – making traffic between two endpoints both private (unreadable to anyone in between) and, often, appear to originate from a different network entirely. The "virtual" part matters: no dedicated physical circuit is built; the privacy and integrity come entirely from cryptography layered on top of ordinary internet connectivity.

1.2 WHAT A VPN ACTUALLY GUARANTEES

- **Confidentiality** – traffic is encrypted, so anyone intercepting it in transit (an ISP, a hostile network, a nosy coffee-shop Wi-Fi) sees only ciphertext.
- **Integrity** – cryptographic authentication ensures traffic wasn't tampered with in transit; a modified packet is detected and rejected.
- **Authentication** – both endpoints cryptographically verify each other's identity before any traffic flows, preventing an attacker from simply pretending to be the other end.

A VPN doesn't guarantee anonymity. It's easy to conflate "encrypted" with "untraceable." A VPN hides traffic content and, for remote-access VPNs, your originating IP from the immediate network you're connected through – but the VPN provider or endpoint itself can still see everything, and traffic analysis, endpoint compromise, or simply logging in with an identifiable account all still identify you.

1.3 TWO BROAD VPN CATEGORIES

This guide covers two very different protocol families used to build that encrypted tunnel: **IPsec** (Sections 2–3), an older, extremely thorough, standards-body-designed suite with enormous flexibility and correspondingly high complexity, and **WireGuard** (Sections 4–5), a modern, deliberately minimal protocol built around a single, opinionated cryptographic design. Both solve the same fundamental problem; they differ enormously in how they get there.

IPsec: Phase 1 (IKE) & Phase 2

2.1 WHY TWO PHASES?

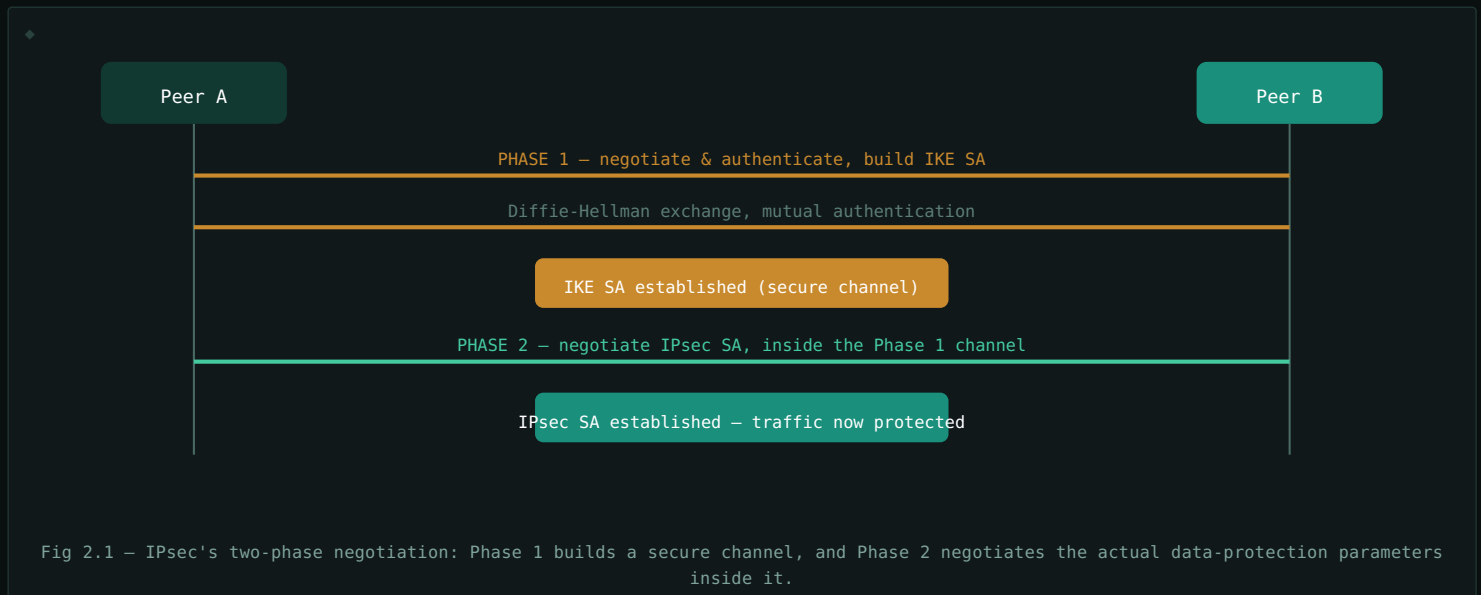
IPsec doesn't jump straight into encrypting data. It first negotiates *how* to securely negotiate everything else – a bootstrapping problem solved by splitting the process into two phases, both driven by the **IKE (Internet Key Exchange)** protocol.

2.2 PHASE 1: BUILDING THE SECURE CHANNEL

Phase 1 establishes an authenticated, encrypted channel called the **IKE SA (Security Association)** – used purely to protect the *negotiation itself*, not yet any real traffic. Both peers authenticate each other (via a pre-shared key or certificates) and agree on encryption/hashing algorithms, typically using a Diffie-Hellman exchange to derive shared keying material without ever transmitting the key itself across the network.

2.3 PHASE 2: NEGOTIATING THE ACTUAL DATA PROTECTION

With Phase 1's secure channel in place, **Phase 2** negotiates the **IPsec SA** – the actual parameters that will protect real user traffic (which encryption algorithm, which mode, which traffic is covered). This negotiation itself now travels safely inside the Phase 1 channel, rather than being exposed in the clear.

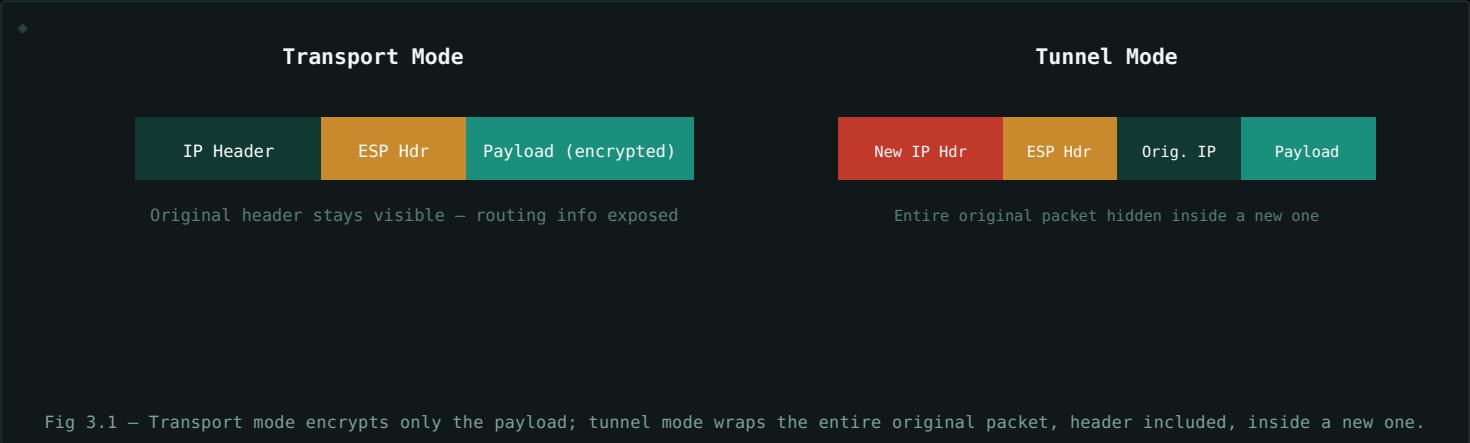


SAs are unidirectional and expire. A Security Association only protects traffic in one direction, so a working IPsec tunnel actually involves a pair of SAs. Both IKE and IPsec SAs also have a limited lifetime and are periodically re-negotiated (called **rekeying**), which limits how much data any single key ever protects.

IPsec Modes: Transport vs Tunnel, AH vs ESP

3.1 TRANSPORT MODE VS TUNNEL MODE

Transport mode encrypts only the payload of the original IP packet, leaving the original IP header intact and visible – used when both endpoints are the actual communicating hosts themselves. **Tunnel mode** encrypts the *entire* original packet, including its header, and wraps it inside a brand-new IP packet with new source/destination addresses – used when IPsec is protecting traffic on behalf of other devices behind it, which is the far more common case for VPNs.



3.2 AH VS ESP: THE TWO PROTECTION PROTOCOLS

AH (Authentication Header) provides integrity and authentication – proof the packet wasn't altered and really came from who it claims – but *no encryption* at all; the payload remains readable. **ESP (Encapsulating Security Payload)** provides both encryption *and* authentication, and is what nearly every real-world IPsec VPN actually uses today. AH has fallen out of practical use largely because it can't function through NAT – its integrity check covers IP header fields that NAT rewrites in transit (see the NAT Deep Dive guide, Section 9, for the full explanation and the NAT-T workaround).

PROTOCOL	ENCRYPTION?	AUTHENTICATION?	WORKS THROUGH NAT?
AH	No	Yes	No – broken by header rewriting
ESP	Yes	Yes	Yes, with NAT-T (UDP encapsulation)

In practice: the overwhelming majority of real IPsec deployments use ESP in tunnel mode – it's the combination that provides full encryption, works through NAT with NAT-T, and correctly hides the original packet's routing information for site-to-site and remote-access VPN traffic alike.

WireGuard: A Modern Approach

4.1 DELIBERATELY MINIMAL

Where IPsec offers extensive negotiable options – multiple encryption algorithms, multiple modes, a two-phase handshake – **WireGuard** takes the opposite philosophy entirely: it hardcodes one modern, carefully chosen cryptographic suite (ChaCha20 for encryption, Curve25519 for key exchange, BLAKE2s for hashing) with no negotiation at all. Fewer options means a dramatically smaller codebase – WireGuard's implementation is often cited at around 4,000 lines of code, versus hundreds of thousands for mature IPsec stacks – which makes it far easier to audit for security vulnerabilities.

4.2 CRYPTOKEY ROUTING: NO CENTRAL SESSION STATE

Instead of IPsec's negotiated Security Associations, WireGuard associates each peer with a public key and a list of allowed IP ranges, directly in a simple configuration – a model called **Cryptokey Routing**. There's no separate "session" concept to negotiate or tear down; a peer is either configured and trusted, or it isn't. This radically simplifies both the protocol and its configuration compared to IPsec's phase negotiation.

ASPECT	IPSEC	WIREGUARD
Cryptographic agility	Many negotiable algorithms	One fixed, modern suite
Codebase size	Very large (mature, feature-rich)	Small (~4,000 lines)
Handshake	Two-phase (IKE Phase 1 & 2)	Single round-trip (Section 5)
Configuration model	Complex policy/SA negotiation	Simple peer + allowed-IPs list
Platform support	Universal, decades of maturity	Growing rapidly, now in mainline Linux kernel

WireGuard is UDP-only and silent by design. An unauthenticated packet gets no response at all – no error, no reset, nothing – making WireGuard servers essentially invisible to port scanners that aren't already carrying a valid key. This is a deliberate security property, not an oversight.

Worked Example: The WireGuard Handshake

WireGuard's handshake is based on the **Noise Protocol Framework** and completes in a single round trip – dramatically simpler than IPsec's multi-message Phase 1 negotiation (Section 2).

1. **Initiator sends a handshake initiation message**, containing an ephemeral public key and an encrypted static public key, to the responder.
2. **Responder verifies the initiator is a known, configured peer** (via Cryptokey Routing, Section 4) and replies with its own handshake response message.
3. **Both sides now independently derive the same symmetric session keys** using the exchanged ephemeral keys – this is the Diffie-Hellman-style exchange that makes the shared key derivable by both ends without ever transmitting the key itself.
4. **Encrypted data can now flow immediately**, using the newly derived session keys – the entire handshake took exactly one round trip.

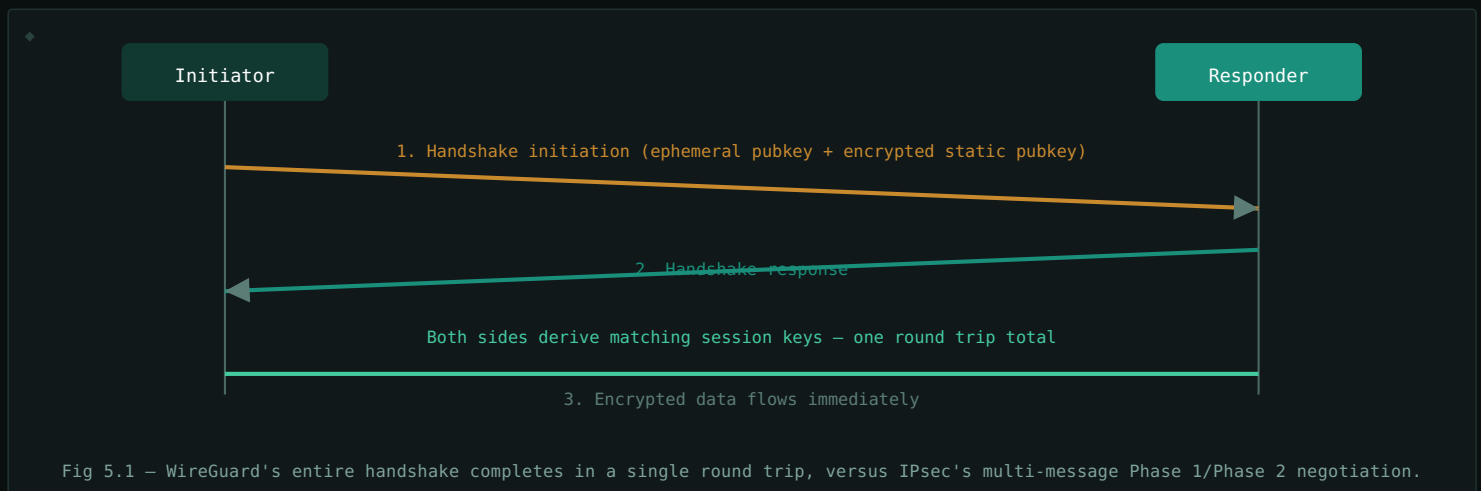


Fig 5.1 – WireGuard's entire handshake completes in a single round trip, versus IPsec's multi-message Phase 1/Phase 2 negotiation.

5.1 AUTOMATIC REKEYING

Like IPsec's SA lifetimes (Section 2), WireGuard automatically re-runs this handshake and derives fresh session keys periodically (by default, every 2 minutes or after a data volume threshold) – limiting how much traffic any single derived key ever protects, without requiring any manual intervention or noticeable interruption to the connection.

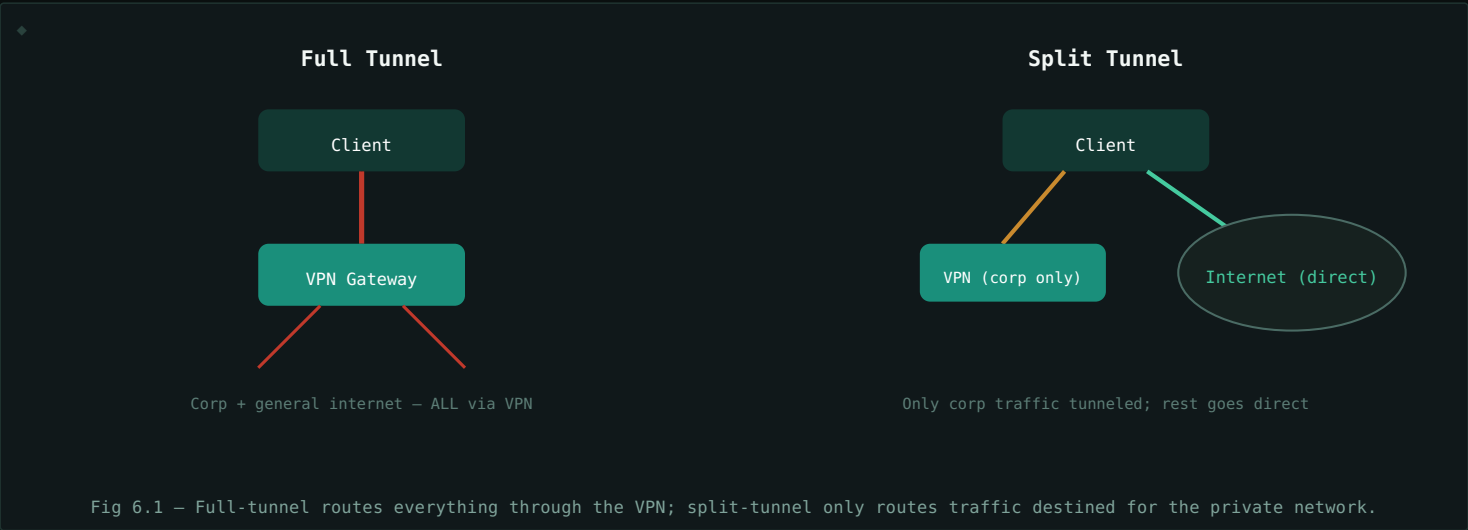
Split-Tunnel vs Full-Tunnel VPNs

6.1 FULL-TUNNEL: EVERYTHING THROUGH THE VPN

In a **full-tunnel** configuration, *all* of a client's internet traffic – not just traffic destined for the private network the VPN connects to – routes through the VPN tunnel. This gives maximum security and control (every packet is inspectable, filterable, and protected by whatever policy governs the VPN's exit point), but adds latency for ordinary internet browsing and consumes significant bandwidth on the VPN concentrator.

6.2 SPLIT-TUNNEL: ONLY SOME TRAFFIC

In a **split-tunnel** configuration, only traffic destined for specific private networks routes through the VPN; everything else (general internet browsing, streaming, etc.) goes directly out the client's own internet connection as normal. This significantly reduces VPN concentrator load and improves performance for non-corporate traffic, at the cost of that traffic never being inspected or protected by the VPN's security policy at all.



ASPECT	FULL-TUNNEL	SPLIT-TUNNEL
Security visibility	All traffic inspectable/filterable	Only corporate-bound traffic covered
Concentrator load	High – carries all client traffic	Lower – only relevant traffic
General browsing performance	Slower – extra hop through VPN	Normal – direct internet path
Common use	High-security environments	General corporate remote access

Site-to-Site vs Remote-Access VPNs

7.1 SITE-TO-SITE: CONNECTING NETWORKS

A **site-to-site VPN** connects two entire networks – typically two office locations, or an office and a cloud VPC (see the upcoming Cloud Networking guide) – through gateway devices at each end. Individual devices on either network have no VPN software or awareness at all; the gateways handle everything transparently, so a user's traffic just flows to the other site as if both networks were directly connected.

7.2 REMOTE-ACCESS: CONNECTING INDIVIDUAL USERS

A **remote-access VPN** connects a single device – a laptop, phone, or individual worker – into a private network from wherever they happen to be. Each client device runs VPN software and authenticates individually, unlike site-to-site's transparent gateway-to-gateway model. This is the classic "VPN client app" most remote workers are familiar with.

ASPECT	SITE-TO-SITE	REMOTE-ACCESS
Connects	Two entire networks	One device to one network
End-device awareness	None – handled entirely by gateways	Requires VPN client software per device
Authentication	Gateway-to-gateway (once, per tunnel)	Per-user, per-device, per-session
Typical use	Office-to-office, office-to-cloud	Remote/traveling employees
Common protocols	IPsec (Sections 2–3)	IPsec, WireGuard, SSL VPN (both common)

These aren't mutually exclusive. A real organization often runs both simultaneously – a site-to-site IPsec tunnel permanently connecting the office to a cloud environment, alongside a remote-access WireGuard or SSL VPN service for individual traveling employees to reach the same private network from their laptops.

Worked Example: Configuring a Site-to-Site VPN

A simplified, vendor-neutral IPsec site-to-site configuration connecting two office gateways, pulling together the Phase 1/Phase 2 concepts from Section 2.

8.1 PHASE 1 (IKE) POLICY

```
$ crypto ike policy 10
$ encryption aes-256
$ hash sha256
$ dh-group 14
$ authentication pre-share
```

Defines the algorithms and authentication method for Phase 1 (Section 2.2) – both gateways must agree on matching values here before the IKE SA can be established.

8.2 PHASE 2 (IPSEC) TRANSFORM SET

```
$ crypto ipsec transform-set VPN-SET esp-aes-256 esp-sha256-hmac
$ mode tunnel
```

Specifies ESP (Section 3.2) with AES-256 encryption in tunnel mode (Section 3.1) – the standard combination for site-to-site VPNs.

8.3 DEFINE INTERESTING TRAFFIC & THE TUNNEL

```
$ access-list VPN-TRAFFIC permit ip 10.1.0.0 0.0.255.255 10.2.0.0 0.0.255.255
$ crypto map SITE-VPN 10 ipsec-isakmp
$ set peer 203.0.113.50
$ set transform-set VPN-SET
$ match address VPN-TRAFFIC
```

The access list defines which traffic should actually be encrypted and sent through the tunnel – traffic between the two offices' private subnets – while everything else continues to route normally, outside the VPN.

In practice: exact syntax varies significantly by platform, and many modern deployments increasingly use route-based VPNs (where a virtual tunnel interface handles routing normally) rather than the older policy-based, access-list-driven model shown here – but the underlying Phase 1/Phase 2 negotiation from Section 2 remains the same either way.

Quick Reference & Glossary

TERM	DEFINITION
VPN	Virtual Private Network – an encrypted tunnel across an untrusted network like the internet.
IPsec	A standards-based VPN protocol suite using a two-phase IKE negotiation.
IKE	Internet Key Exchange – the protocol driving IPsec's Phase 1 and Phase 2 negotiation.
Security Association (SA)	A unidirectional, negotiated set of parameters protecting traffic between two IPsec peers.
AH	Authentication Header – provides integrity/authentication only, no encryption; broken by NAT.
ESP	Encapsulating Security Payload – provides both encryption and authentication; the modern standard.
Transport Mode	IPsec mode encrypting only the payload, leaving the original IP header visible.
Tunnel Mode	IPsec mode encrypting the entire original packet inside a new one.
WireGuard	A modern, minimal VPN protocol with a fixed cryptographic suite and single round-trip handshake.
Cryptokey Routing	WireGuard's model associating each peer with a public key and allowed IP ranges.
Full-Tunnel	A VPN mode routing all client traffic through the tunnel.
Split-Tunnel	A VPN mode routing only traffic destined for the private network through the tunnel.
Site-to-Site VPN	A VPN connecting two entire networks via gateway devices, transparent to end users.
Remote-Access VPN	A VPN connecting an individual device into a private network, requiring client software.

End of guide. IPsec and WireGuard solve the same problem from opposite design philosophies – exhaustive flexibility versus deliberate minimalism – and which one fits depends far more on your environment's existing infrastructure and platform support than on any inherent superiority of one over the other.